A Unified Theory of Geological Resources, Dynamic Pricing, Costs, State Capacity, and Strategic Power

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Abstract

This paper unifies two previously separate frameworks:

1. a heterogeneous-Earth resource economy possessing a radially varying geological density field;
2. a structurally closed monetary–geopolitical manifold governing dynamic pricing and strategic interactions.

An embedding map transports geological locations into monetary–geopolitical state space. Prices, costs, margins, and effective resource densities become manifold-valued fields. The resulting profit functional is shown to be the pullback of manifold fields onto the geological domain. The framework integrates geology, economics, finance, political science, international relations, warfare economics, statistics, computer science, machine learning, and dynamic optimization into a unified mathematical architecture.

The paper ends with "The End"

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1 Introduction

Traditional resource economics typically treats geological resources and market prices as separate objects.

Geology determines resource abundance.

Markets determine prices.

Political systems determine institutions.

Geopolitics determines strategic constraints.

In reality, these systems interact continuously.

The objective of this paper is to unify:

$$Geology \to \begin{matrix} Resource \\ Extraction \end{matrix} \to \begin{matrix} Monetary \\ Valuation \end{matrix} \to \begin{matrix} State \\ Capacity \end{matrix} \to Geopolitics \to Prices\ and\ Costs.$$

The result is a closed feedback system operating on a monetary–geopolitical manifold.

2 Background

This paper builds upon two earlier frameworks.

2.1 The Heterogeneous-Earth Resource Economy

The heterogeneous-Earth model introduced:

$$\rho(r),$$

a radially varying density field satisfying

$$M = \int_0^R 4\pi\rho(r)r^2 dr.$$

Extractable mass was

$$M_{\text{ext}} = \int_{r_{\max}}^R 4\pi\rho(r)r^2 dr.$$

Profit was

$$\Pi = \int_{r_{\max}}^R 4\pi[p(r) - c(r)]\rho(r)r^2 dr.$$

2.2 The Monetary–Geopolitical Manifold

The monetary–geopolitical framework introduced a state space

$$\mathcal{M},$$

containing monetary, geopolitical, strategic, institutional, and technological coordinates.

Dynamic pricing was represented by

$$p(x,t), \qquad x \in \mathcal{M}.$$

3 The Monetary–Geopolitical Manifold

Definition 1. *A structurally closed monetary–geopolitical manifold is a smooth manifold*

$$\mathcal{M}$$

whose coordinates describe the state of a political-economic system.

Typical coordinates include

$$x = (m, g, s, \tau, \ell, \kappa, \eta, \ldots),$$

where

- m = monetary variables,
- g = geopolitical variables,
- s = strategic variables,
- τ = institutional variables,
- ℓ = logistics,
- κ = technological capability,
- η = military capability.

4 Geological Embedding

Definition 2. *Define the embedding*

$$\Phi : [0, R] \to \mathcal{M}.$$

Each geological radius is mapped into the manifold:

$$r \mapsto x(r).$$

Thus every geological location possesses a corresponding monetary–geopolitical state.

Remark 1. *The embedding converts a physical Earth into an economically interpretable object.*

5 Manifold-Valued Price and Cost Fields

5.1 Price Field

Define

$$p(x, t).$$

The induced geological price field becomes

$$p(r, t) = p(x(r), t).$$

5.2 Cost Field

Define

$$c(x,t).$$

The induced geological cost field becomes

$$c(r,t) = c(x(r),t).$$

Unlike traditional extraction models, costs are not purely engineering quantities. They depend upon:

- interest rates,
- wages,
- sanctions,
- insurance,
- military risk,
- technological capability,
- regulation.

6 Margin Field

Definition 3. *Define the manifold margin field*

$$m(x,t) = p(x,t) - c(x,t).$$

The geological margin becomes

$$m(r,t) = m(x(r),t).$$

7 Effective Density Fields

Physical density alone is insufficient.

Recoverable density depends upon:

- technology,
- regulation,
- ownership rights,
- military control,
- environmental restrictions.

Therefore define

$$\rho(x,t).$$

The induced density field becomes

$$\rho(r,t) = \rho(x(r),t).$$

8 The Unified Profit Functional

Theorem 1 (Unified Profit Functional)**.** *Profit is*

$$\boxed{\Pi = \int_{r_{\max}}^{R} 4\pi m(x(r),t)\rho(x(r),t)r^2 dr.}$$

Proof. Substitute

$$m(x,t) = p(x,t) - c(x,t)$$

into

$$\Pi = \int_{r_{\max}}^{R} 4\pi[p-c]\rho r^2 dr.$$

The result follows immediately.

□

9 Pullback Interpretation

Let

$$\Phi^* m$$

denote the pullback of the margin field.
Similarly define

$$\Phi^* \rho.$$

Then

$$\Pi = 4\pi \int_{r_{\max}}^{R} (\Phi^* m)(\Phi^* \rho) r^2 dr.$$

This formulation is geometrically natural.

10 Closed Feedback Dynamics

Resource wealth equals

$$W(t) = \Pi(t).$$

Suppose manifold evolution satisfies

$$\dot{x} = F(x, W).$$

Then

$$W \to x \to m \to W.$$

This produces a structurally closed dynamical system.

11 State Capacity

Let

$$S(t)$$

denote state capacity.
Assume

$$S(t) = S_0 + \alpha W(t).$$

Then

$$\dot{x} = F(x, S).$$

Hence resource extraction affects state power.

12 Geopolitical Power

Let

$$G(t)$$

denote geopolitical influence.
Suppose

$$G(t) = G_0 + \beta S(t).$$

Then

$$G(t) = G_0 + \beta S_0 + \alpha\beta W(t).$$

Resource wealth therefore influences geopolitical influence.

13 International Relations

Suppose multiple states inhabit the manifold.
Let

$$x_i \in \mathcal{M}.$$

Interactions satisfy

$$\dot{x}_i = F_i(x_1, \ldots, x_n).$$

The manifold naturally accommodates:

- alliances,
- trade networks,
- sanctions,
- strategic competition.

14 Warfare Economics

Military power satisfies

$$M_S = M_0 + \gamma W.$$

Military expenditure is

$$D = \delta W.$$

Resource extraction therefore finances military capability.

15 Welfare Economics

Social welfare is

$$U(C) - D(E),$$

where

- C = consumption,
- E = environmental damage.

The planner solves

$$\max \Big(U(C) - D(E) \Big).$$

16 Finance

Resource wealth generates financial assets.

Firm valuation becomes

$$F = \sum_{t=0}^{\infty} \frac{E[\Pi_t]}{(1+i)^t}.$$

Resource firms become manifold-dependent claims.

17 Psychology and Psychiatry

Expectations influence prices:

$$E_t[p_{t+1}].$$

Behavioral biases affect

- valuation,
- risk assessment,
- investment.

Resource shocks may influence stress, migration, and social stability.

18 Physics

Extraction requires energy.

Define

$$\varepsilon(x, t).$$

Total energy requirement is

$$E_{\text{req}} = \int_{r_{\max}}^{R} 4\pi\varepsilon(x(r), t)\rho(x(r), t)r^2 dr.$$

19 Chemistry

Resource concentration field:

$$\chi(x, t).$$

Recoverable resource quantity:

$$Q = \int_{r_{\max}}^{R} 4\pi\chi(x(r), t)\rho(x(r), t)r^2 dr.$$

20 Biology

Environmental impacts influence biological systems.

Let biodiversity be

$$B(t).$$

Assume

$$\frac{dB}{dE} < 0.$$

Resource extraction therefore interacts with ecological systems.

21 Statistics

Expected profit:

$$E[\Pi] = \int_{r_{\max}}^{R} 4\pi E[m(x(r), t)\rho(x(r), t)]r^2 dr.$$

Variance:

$$\text{Var}(\Pi) = E[\Pi^2] - (E[\Pi])^2.$$

22 Machine Learning

A predictor learns

$$Y = f_\theta(X).$$

Examples:

- geological forecasting,
- price forecasting,
- conflict forecasting.

23 Reinforcement Learning

State:

$$s_t.$$

Action:

$$a_t.$$

Reward:

$$R_t = \Pi_t.$$

Objective:

$$\max_{\pi} E\left[\sum_{t=0}^{\infty} \beta^t R_t\right].$$

24 World Models

The manifold evolves according to

$$x_{t+1} = F(x_t, a_t, \varepsilon_t).$$

World models simulate future trajectories.

25 Equilibrium

Theorem 2 (Monetary–Geopolitical Resource Equilibrium)**.** *Suppose there exists*

$$x^* \in \mathcal{M}$$

such that

$$x^* = F(x^*, \Pi^*).$$

Suppose

$$\Pi^* = \int_{r_{\max}}^{R} 4\pi m(x^*)\rho(x^*)r^2 dr.$$

Then

$$(x^*, \Pi^*)$$

constitutes a manifold equilibrium.

Proof. At equilibrium,

$$\dot{x} = 0.$$

Hence

$$x^* = F(x^*, \Pi^*).$$

Substituting into the profit functional yields the fixed-point condition.

□

26 A Unified Geological–Monetary–Geopolitical Field Theory

The complete system is

$$\boxed{\Pi = \int_{r_{\max}}^{R} 4\pi m(x(r), t)\rho(x(r), t) r^2 dr.}$$

$$\boxed{\dot{x} = F(x, \Pi).}$$

Together these equations define a coupled geological–monetary–geopolitical field theory.

27 Summary Table

Symbol	Interpretation
$\mathcal{M}$	Monetary–geopolitical manifold
Φ	Geological embedding
$x(r)$	Embedded manifold coordinate
$p(x,t)$	Price field
$c(x,t)$	Cost field
$m(x,t)$	Margin field
$\rho(x,t)$	Effective density field
Π	Profit functional
W	Resource wealth
S	State capacity
G	Geopolitical power
M_S	Military capability

Table 1: Core variables of the unified framework.

28 Illustration of the Embedding

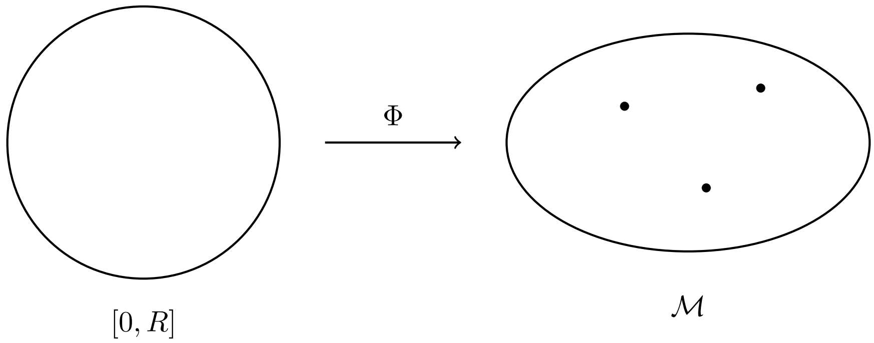

Figure 1: Illustration of the Embedding.

References

[1] Ghosh, S. *A Stochastic Resource-Extraction Economy of Earth with Heterogeneous Geology*. Unpublished manuscript.

[2] Ghosh, S. *Dynamic Pricing on a Structurally Closed Monetary–Geopolitical Manifold*. Collected Papers, Book 121.

[3] Hotelling, H. The Economics of Exhaustible Resources. *Journal of Political Economy*, 1931.

[4] Arrow, K. J. *Social Choice and Individual Values*. Yale University Press.

[5] Sutton, R. S. and Barto, A. G. *Reinforcement Learning: An Introduction*. MIT Press.

[6] von Neumann, J. and Morgenstern, O. *Theory of Games and Economic Behavior*. Princeton University Press.

Glossary

$\mathcal{M}$ Monetary–geopolitical manifold.

Φ Embedding from geological space into manifold space.

$x(r)$ Manifold image of radius r.

$p(x, t)$ Price field.

$c(x, t)$ Cost field.

$m(x, t)$ Margin field.

$\rho(x, t)$ Effective density field.

W Resource wealth.

S State capacity.

G Geopolitical power.

Π Profit functional.

The End